



**Vilnius
University**



Internal IT Helpdesk System Project

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Project Introduction

Project Topic and Relevance

This project focuses on the analysis and improvement of an Internal IT Helpdesk System for a university, including features such as ticketing, request tracking, knowledge base management, and automated workflows. As universities continue to expand their digital infrastructure supporting online learning platforms, campus WiFi, computer labs, administrative portals, cybersecurity systems, and cloud-based services the need for efficient IT support becomes increasingly critical. A modern helpdesk system enables students, academic staff, and administrative departments to report issues, request services, and access self-service information quickly and reliably.

Business / IT Systems Context

- **Operational Efficiency:** Universities rely on complex IT infrastructure lecture capture systems, LMS platforms, labs, networks, and administrative tools. A helpdesk system ensures timely resolution of incidents and service requests, reducing disruptions.
- **User Satisfaction:** Staff and students depend on reliable IT services. An effective helpdesk improves overall user experience, trust, and productivity.
- **Resource Optimization:** Helps IT departments allocate resources based on real demand and analytics (e.g., recurring issues, peak request times).
- **Strategic Alignment:** A helpdesk supports the university's digital transformation goals by improving service management maturity.
- **Knowledge Management:** Captures organizational knowledge via documentation, ticket history, and common solutions, enabling better long-term IT planning.
- **Compliance & Governance:** Structured ticketing enables auditability, SLA tracking, and adherence to ITIL-based best practices.

Objectives and Tasks

To analyze how the Internal IT Helpdesk System supports the university's strategic goals for digital transformation and identify where improvement opportunities lie.

Tasks:

1. Perform PESTLE and VMOST analysis to evaluate the strategic context and external factors affecting IT support at the university.
2. Evaluate stakeholder positions using Stakeholder Wheel, CATWOE, and Business Activity Model (BAM) to understand roles, influences, and expectations.
3. Assess existing IT support processes with frameworks such as McKinsey 7S to identify current weaknesses and improvement areas.
4. Propose a solution for the Internal IT Helpdesk using POPIT perspectives, focusing on processes, people, information, and technology.
5. Evaluate the expected value of the solution through a Benefits Dependency Network and CARDI analysis to justify investment and highlight potential benefits and risks.
6. Define key requirements for the new helpdesk system, covering functional, non-functional, technical, and legal aspects, along with validation and test scenarios.

Research Methods

To ensure a comprehensive understanding of the existing situation and the design of an effective Internal IT Helpdesk System, several research and analysis methods are used throughout this project:

- Interviews: Semi-structured interviews with IT support staff, the IT manager, and representatives from students, faculty, and administrative departments to identify pain points, expectations, and improvement suggestions regarding IT support.
- Process analysis: Observation and documentation of current IT support processes (e.g., handling requests via email or phone) to identify inefficiencies, bottlenecks, and informal workarounds.
- Persona analysis: Creation of representative user personas (e.g., “undergraduate student”, “lecturer”, “IT agent”) to capture different needs, behaviors, and service requirements relevant to the helpdesk system.
- Document and data review: Analysis of existing incident logs, ticket records (if available), and IT policies to map common issue categories, response times, and service gaps.
- Modeling: Use of established business analysis frameworks (PESTLE, VMOST, McKinsey 7S, Stakeholder Wheel, CATWOE, BAM, POPIT, CPPOLDAT) to structure findings and proposed solutions.

Strategic Context

Organization Overview

The case organization is a mid-sized university with approximately 10,000 students, 600 academic staff, and 300 administrative employees.

The university is currently considered to be a good regional academic institution that serves its market using basic digitalization tools. The university is widely acknowledged for its teaching, research, and community engagement and its core assets include distinguished academic courses, a highly qualified faculty, and a foundation of well-established campus facilities.

The target state of the university is to become a digitally empowered, student-centered, and globally recognized institution committed to supporting modern learning and research methods through advanced digital technologies.

Current Situation

While the university has successfully adopted various digital tools (LMS, cloud storage, hybrid learning), its operational support structure has failed to keep pace with this digital expansion. As the dependency on technology has increased, the volume of support requests from students and faculty has surged. Currently, there is no centralized mechanism to manage this demand. The

support process is informal and fragmented, relying on walk-ins, direct emails to technicians, and phone calls. This disconnect creates a bottleneck where strategic goals (like digital empowerment) are stifled by operational inefficiencies, leading to dissatisfied stakeholders and wasted staff hours.

POPIT – Strategic Context for the Internal IT Helpdesk System

People:

- Students, academic staff, and administrative employees increasingly depend on digital services (LMS, cloud tools, virtual classrooms) for everyday learning, teaching, and administration.
- IT support staff are under pressure to handle growing volumes of requests with limited structure, leading to stress, overload, and inconsistent service quality.

Organization:

- The university is a reputable regional institution with strong teaching and research reputation but operates with basic, partially digitalized support processes.
- Its strategic aim is to become a digitally empowered, student-centered and globally recognized university, yet the lack of a formal support structure creates a gap between strategy and day-to-day operations.

Process:

- Current support processes are informal and fragmented: walk-ins, direct emails, and phone calls with no standard workflow, prioritization, or tracking.
- This leads to duplicated work, unresolved or forgotten requests, slow response times, and no reliable performance indicators to support strategic decision-making.

Information:

- Information about incidents and service requests is scattered across personal inboxes, ad-hoc notes, and verbal agreements, making it impossible to obtain a complete view of demand or recurring issues.
- The absence of centralized, structured data prevents the university from learning from past cases, building a reusable knowledge base, or aligning IT improvements with strategic priorities.

Technology:

- Although the university has invested in learning platforms, networks, and cloud tools, there is no dedicated technology to coordinate and monitor IT support activities.

- Implementing an Internal IT Helpdesk System would introduce a centralized platform for ticketing, tracking, knowledge management, and analytics, aligning the operational support layer with the university's broader digital transformation goals.

Proposed IT solution

The organization plans to implement a centralized Internal IT Helpdesk System. This solution will replace the current ad-hoc communication methods with a structured platform featuring:

1. **Ticketing and Request Tracking** to capture, prioritize, and assign issues centrally.
2. **Implementing a knowledge base** to allow users to resolve common issues independently.
3. **Automating routine processes** such as ticket routing, notifications, and SLA monitoring.
4. **Monitoring performance** through analytics and feedback to provide management with data on response times, common bottlenecks and guide continuous enhancement.

Topic's importance to the organization

This topic is critical because the current inefficiency is a direct risk to the university's strategic vision of "digital empowerment".

1. **Student Experience & Retention:** In a competitive higher-education market, students expect seamless digital experiences. Long wait times for IT support negatively impact their learning experience and satisfaction.
2. **Operational Efficiency:** Highly paid academic staff are currently wasting time chasing technical support, reducing their capacity for teaching and research.
3. **Data-Driven Decision Making:** Without a system to track issues, leadership lacks the data to know where to invest in training or infrastructure upgrades. The new system will provide the "rational basis" for future IT strategy.

PESTLE Analysis

A PESTLE analysis highlights the external factors influencing the need for an IT Helpdesk System.

Political

- Increased government emphasis on digital education and cybersecurity.
- Public-sector universities often face audits requiring documented IT service processes.

Economic

- Budget constraints require cost-effective IT resource allocation. Efficiency gains through automation are increasingly necessary due to rising student numbers.

Social

- Students expect fast, accessible, 24/7 digital support.
- Hybrid learning models create higher dependency on technology.

Technological

- Growth in cloud computing, automation, and AI-based support tools.
- Increasing system complexity requires centralized ticket management and knowledge sharing.

Legal

- GDPR and data protection regulations require secure handling of personal information.
- Requirements for auditable logs and traceable decision-making processes.
- The education sector is a prime target for cyberattacks (e.g., phishing). A fragmented support system via email is insecure and makes it difficult to track patterns of security incidents. A centralized system allows for better incident logging and trend analysis.

Environmental

- Reduction of paper-based processes contributes to sustainability goals.
- Remote support reduces unnecessary travel for campus IT teams.

PESTLE Conclusion:

External pressures, especially technological change, legal compliance, and student expectations strongly justify the implementation of a structured helpdesk system.

Strategic Alignment (VMOST Mapping)

V MOST Element	University Strategic Aims / Elements	IT Helpdesk System Alignment / Contribution
Vision	Become a leading, digitally empowered and student-centered university, recognized for excellence in teaching and IT.	The helpdesk system supports digital transformation by ensuring robust, reliable IT services for all academic operations.
Mission	Deliver seamless, high-quality, and efficient IT support for all users (students, faculty, staff).	Centralized ticketing and knowledge base improve service speed, transparency, and user satisfaction.
Objectives	Reduce the average IT issue resolution time by 30% within 12 months. Enhance user satisfaction by 20%.	Automated workflows, SLA monitoring, and analytics drive measurable improvement in ticket resolution and satisfaction.
Strategy	Implement modern, consolidated IT management tools and processes across campus.	The helpdesk replaces fragmented support channels with integrated end-to-end ticket management, knowledge sharing, and analytics.
Tactics	Roll out helpdesk portal, train IT staff, promote self-service for common issues, monitor KPIs.	Features include ticket categorization, automated routing, self-service knowledge base, real-time performance reporting.

Stakeholder Analysis

Stakeholder Identification and Mapping

The Internal IT Helpdesk System affects several key stakeholder groups in the university:

- **University board / senior leadership:** Set overall strategy and approve budgets for major IT initiatives.
- **Project manager and helpdesk manager:** Plan, coordinate, and oversee the implementation and daily operation of the helpdesk system.
- **Department heads:** Lead academic or administrative units and influence how IT policies and processes are adopted locally.
- **IT team:** Includes helpdesk agents, technicians, and system administrators who handle incidents, requests, and maintenance tasks using the new platform.
- **Administrative staff:** Use student information, finance, HR, and other systems and rely on stable IT services for daily work.
- **Faculty staff:** Lecturers and researchers who depend on reliable digital services to deliver courses and conduct research.
- **Students:** The largest user group, heavily dependent on LMS, Wi-Fi, labs, and other IT services, and therefore frequent requestors of IT support.
- **External vendors:** Providers of specialised systems (e.g., LMS, network, cloud services) who support incident resolution and system integrations when needed.

These stakeholders form an interconnected ecosystem: users raise issues, IT staff resolve them, management monitors performance, leadership validates the value, and vendors contribute to solving complex or external problems.

Influence and Interest (Power/Interest Matrix)

Stakeholders can be positioned on a Power/Interest Matrix to guide engagement:

Stakeholder	Power / Influence	Interest
University Board	High	Low
Project Manager	High	High
Helpdesk Manager	High	High

Department Heads	Some	Some
IT Team	Some	High
Administrative Staff	Some	Low
Faculty Staff	Low	Some
Students	Low	Low
Vendors	Low	Low

This mapping helps prioritise communication and change-management efforts, increasing the likelihood of system adoption and long-term success.

RASCI Matrix and CATWOE Analysis

Project Activity / Deliverable	Project Manager	Business Analyst	Uni Board	Helpdesk Manager	IT Team	Faculty Members / Students	Vendor
1. Define Business Requirements (Gathering needs for ticketing & knowledge base)	A	R	I	C	C	C	C
2. Select IT Solution (Choosing the software vendor)	A	S	C	R	C	I	C
3. Design "To-Be" Process (How a ticket	I	R	I	A	C	I	I

flows from open to close)							
4. Configure Automation Rules (e.g., auto-assigning password resets)	I	S	C	A	R	I	I
5. User Acceptance Testing (UAT) (Verifying the system works)	I	A	I	S	R	I	S
6. Data Migration (Moving old emails/logs to new system)	I	I	I	A	R	I	I
7. Training & Go-Live (Launch to the University)	A	S	I	R	S	I	S

CATWOE perspective for the helpdesk system:

	Perspective of the Helpdesk Manager	Perspective of the Student
W	Operational Stability & Efficiency	Accessibility & Enabler
T	Unmanaged incidents → Resolved & logged tickets	Inability to study/work → Ability to continue work
C	The University Community (Students, Faculty, and Staff)	The Student
A	IT Support Teams & Vendors	The IT Helpdesk System &

		Staff
O	Chief Information Officer (CIO)	University Administration
E	Operational Constraints	Academic Constraints

Business Activity Model

The following unified Business Activity Model consolidates activities across key stakeholders into a single, aligned view, supporting backlog creation and operational process design.

Activity Type	Business Activities
Doing	• Log and categorise IT incidents and service requests through the helpdesk system. • Assign tickets to the appropriate support group or agent based on category and priority. • Diagnose and resolve incidents or fulfil service requests according to defined procedures. • Communicate with users regarding ticket progress, clarification requests, and resolution. • Close tickets after resolution and record resolution details.
Enabling	• Maintain the helpdesk system, user access rights, and integrations (SSO, email). • Train IT staff and users on helpdesk processes and system usage. • Develop and maintain a knowledge base with reusable solutions and FAQs. • Provide user guidelines for correct ticket submission and self-service usage.
Planning	• Define IT support policies, service catalogue, and Service Level Agreements (SLAs). • Plan resource allocation and staffing levels based on expected demand (e.g. exam periods). • Prioritise improvement initiatives based on recurring incidents and system analytics.
Monitoring	• Monitor ticket queues, resolution times, and SLA compliance in real

	time. • Track key performance indicators (KPIs) such as First Contact Resolution and user satisfaction. • Review trends in incident categories and request volumes.
Control	• Reassign tickets or resources to address bottlenecks or SLA risks. • Escalate major incidents to senior management or external vendors when required. • Implement process and system improvements based on performance reviews and user feedback.

Process Analysis

As-is Process Description

Currently, IT support at the university is handled through informal and fragmented channels. Users contact IT staff via personal emails, walk-in visits to the IT office, or direct phone calls to individual technicians. There is no centralized tool for logging, categorizing, or prioritizing requests, so incidents and service requests are tracked inconsistently and often depend on personal memory or ad-hoc notes.

As a result, multiple issues arise. Requests can be overlooked or significantly delayed, as there is no single queue or overview of all open tickets. Different technicians may duplicate work if the same problem is reported through separate channels. Users have limited visibility into the status of their issues and frequently need to follow up manually, increasing frustration. IT management cannot easily measure workload, resolution times, or common types of incidents, which makes it difficult to allocate resources, justify staffing levels, or identify areas for improvement.

To-be Process Proposal

In the proposed future state, all IT incidents and service requests will be managed through a centralized Internal IT Helpdesk System. Users (students, academic staff, and administrative staff) will submit requests via an online portal or integrated intranet form, where they can describe the issue, select a category, attach screenshots or files, and indicate urgency. Each request is automatically recorded as a ticket with a unique ID.

The helpdesk system will apply predefined rules to categorize and route tickets to the appropriate support queue (e.g. network, accounts, hardware). IT agents will work from a shared ticket dashboard, where they can see priorities, deadlines (SLAs), and ownership for each case. Users will be able to track the status of their tickets in real time and receive automated notifications when the ticket is updated or resolved. After resolution, agents can add solutions to a knowledge base, enabling self-service for similar issues in the future. IT management will access reports on ticket volumes, resolution times, SLA compliance, and recurring problems to guide process improvements and strategic decisions.

Improvements

- Centralized and transparent handling of all IT requests, with a single source of truth instead of fragmented emails and phone calls.
- Faster and more predictable resolution times through prioritization, routing rules, and SLA monitoring.
- Reduced duplicate work and fewer “lost” requests, as every incident is tracked from creation to closure.
- Better user experience, because students and staff can see ticket status, receive notifications, and use self-service articles.
- Data-driven decision-making for IT management, based on reports about workloads, common issues, and performance trends.

Process Evaluation Models (e.g., McKinsey 7S)

To evaluate the organisational readiness for the new helpdesk process, the McKinsey 7S framework can be applied:

- Strategy:
The university’s strategy focuses on digital transformation and student-centred services. Implementing a structured helpdesk process supports this by ensuring reliable IT operations and measurable service quality.
- Structure:
Currently, IT support is organised informally with loose role boundaries. The new process requires clearer structure, with defined first-line and second-line support roles and responsibility for ticket queues.
- Systems:
Existing “systems” are mainly email, phone, and informal spreadsheets. The

proposed helpdesk introduces a formal system for ticket management, knowledge base, and reporting, which becomes a core operational backbone for IT support.

- Shared Values:

The university promotes quality teaching, research, and service to students. A professional helpdesk reinforces values of responsiveness, transparency, and continuous improvement in IT support.

- Skills:

IT staff possess strong technical skills but limited experience with formal service management practices. Training in ticketing workflows, communication, and use of the helpdesk tool will be necessary to maximize benefits.

- Style:

The current support style is reactive and person-based (users contact “someone they know”). The target style is more process-oriented and collaborative, where issues are handled through a shared queue with clear ownership and service standards.

- Staff:

The number of IT staff is limited, and they already manage a high workload. The new process and system will help them work more efficiently by reducing duplicate work, enabling better prioritisation, and supporting data-driven staffing decisions.

Solution Design

Gap Analysis

The current IT support process is informal, fragmented, and difficult to manage, whereas the target state requires a structured, data-driven helpdesk capability. The main gaps are:

- **Process gap:** There is no standardized workflow for logging, classifying, prioritizing, and resolving requests. The target state requires a clearly defined ticket lifecycle with ownership, SLAs, and escalation paths.
- **Information gap:** Incident and request data are scattered across inboxes and verbal agreements. The future solution needs a centralized repository of ticket data and a searchable knowledge base for recurring issues.
- **Technology gap:** At present, there is no dedicated helpdesk platform; support relies on generic tools such as email and phone. The target state introduces an integrated helpdesk system with ticketing, automation, and reporting.

- **Management gap**: IT management lacks reliable metrics on volumes, response times, and problem trends. The new solution must provide dashboards and reports that support planning, staffing, and continuous improvement.
- **User experience gap**: Users do not have a consistent way to request support or track progress. The target solution offers a self-service portal, transparent status tracking, and access to help articles.

The Internal IT Helpdesk System is designed to close these gaps by standardising workflows, centralising information, providing appropriate technology support, and improving both management insight and user experience.

POPIT Analysis

People

- IT staff will move from ad-hoc handling of issues to working through a shared ticketing platform with defined responsibilities and priorities. Training will be required to familiarise them with the new workflows, SLAs, and communication standards.
- Students, academic staff, and administrative staff will be educated to use the portal and knowledge base instead of informal channels, increasing self-reliance and reducing frustration.

Organisation

- The IT function will operate with clearer roles (first-line, second-line support, supervisors) and documented procedures.
- The helpdesk will support the university's strategic objectives by providing a formal mechanism to deliver and measure digital support services aligned with institutional goals.

Process

- A standard end-to-end process will be established: request logging, categorisation, prioritisation, assignment, resolution, communication, and closure.
- Integration with change management and problem management processes can be introduced later, enabling root-cause analysis and controlled deployment of fixes.

Information

- All incidents and requests will be recorded in a single system, providing a complete history for each user and asset.

- A structured knowledge base will capture solutions to frequent issues, which can be reused by IT staff and offered as self-service content for users.

Technology

- The university will adopt an Internal IT Helpdesk platform that supports ticketing, automation rules, knowledge base, SLA tracking, and reporting.
- The system should integrate with existing authentication (e.g., single sign-on) and core university systems where necessary (e.g., LMS, identity management).

CPPOLDAT Analysis

Customers

- All main customer groups (students, academic staff, administrative staff, IT management, university leadership) have been identified and are aware of the upcoming Internal IT Helpdesk System and the related changes in the support process.

Process

- The end-to-end helpdesk process (ticket creation, categorisation, assignment, diagnosis, resolution, user communication, and closure with feedback) is documented at a **high level** and aligned with the university's IT architecture and digital-transformation strategy. This ensures processes are ready for implementation while maintaining flexibility for adjustments during rollout.

Data

- Ticket, user and knowledge-base data structures are defined, data retention and protection rules comply with GDPR and university policies, and any required data migration from legacy sources has been assessed.

Applications

- A helpdesk/ticketing application supporting ticket metadata, SLA tracking, knowledge base and reporting has been selected, with confirmed integrations to SSO, LMS and email for notifications.

Technology

- The underlying infrastructure (hosting/SaaS environment, HTTPS security, authentication, monitoring) is available and meets the university's technical and security standards.

Organization

- Not applicable

Location

- Not applicable

Value Analysis and Business Case

Benefits and Risks (RAID)

Element	Description	Potential Impact	Mitigation / Management Action
Risk	Users continue to use email/phone instead of the helpdesk portal.	Medium – reduced benefits, fragmented data.	Clear policy, deny response through this channel, communication campaign, and hands-on user training.
Risk	Integration problems with LMS, SSO or email systems.	High – delays go-live, possible data inconsistency.	Early technical assessment, pilot phase, dedicated integration tests.
Assumption	IT staff can allocate time for configuration, testing, and training.	Medium – if false, project delays or lower quality.	Adjust workload, phased rollout, or temporary extra support.
Assumption	Senior management maintains sponsorship and budget.	High – if withdrawn, scope reduction or halt.	Regular progress reports and strong link to digital strategy.
Issue	No structured historical ticket data for realistic SLAs.	Low – initial targets may be inaccurate.	Use the first 3–6 months of data to refine SLAs and targets.
Issue	Existing user frustration with slow IT support.	Medium – scepticism towards the new system.	Involve user reps in design, highlight expected improvements.
Dependency	Vendor reliability for support, updates, and bug fixes.	High – affects stability and roadmap.	Contractual SLAs, escalation paths, vendor performance reviews.

Dependency	Accurate organisational/user data for routing rules and access rights.	Medium – mis-routed tickets and confusion.	Clean and synchronise HR/student data before implementation.
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Benefit Definition

The Internal IT Helpdesk System is expected to deliver the following specific business benefits:

- Time savings
 - Reduced average resolution time for incidents and requests through prioritisation, routing rules, and SLA monitoring.
- Increased user satisfaction
 - Faster responses, transparent ticket status, and self-service knowledge base for students, academic staff, and administrative staff.
- Error reduction and process consistency
 - Standardised workflows and reusable solutions minimise inconsistent handling and repeated mistakes.
- Resource and cost optimisation
 - Visibility into ticket volumes and recurring issues supports more efficient staffing, targeted training, and smarter IT investments.

Benefit Dependency Network

Enabling Changes (Enablers)	Business Changes (Process / Behaviour)	Business Benefits
Implementation of a central helpdesk platform with ticketing and SLA features.	All IT requests are logged, categorised and prioritised in that system.	Shorter average resolution times, fewer lost or duplicated requests.
Introduction of a self-service knowledge base.	Users first search for solutions and only raise tickets when needed.	Reduced workload for IT staff, faster resolution of common issues.
Configuration of routing rules, queues and escalation paths.	Tickets are automatically assigned to the right team with clear ownership and deadlines.	Higher SLA compliance and more predictable service levels.

Regular performance reporting on volume, time, and satisfaction KPIs.	Management reviews data, identifies bottlenecks, and adjusts staffing or processes.	Better resource allocation and continuous process improvement.
Training and communication for students, faculty, and staff on portal and KB usage.	Users adopt the new process, submit better-quality tickets, and rely less on ad-hoc email/phone.	Higher user satisfaction and more efficient communication.

Requirements Management

Requirements List (Functional, Non-functional, Technical, Legal)

General Requirements

- The system must comply with GDPR when storing and processing personal data of students and staff.
- The helpdesk portal must be available in at least the main teaching language of the university.

Functional Requirements

- The system shall allow users to submit IT tickets via a web portal, including category, description, urgency and attachments.

Acceptance criteria:

1. Given the user is logged in, when they open the helpdesk portal, they can access a ticket submission form with fields for category, description, urgency and file attachments.
 2. Given the ticket submission form is open and all mandatory fields are filled, when the user submits the form, the system creates a ticket with a unique ticket ID and stores all entered data.
- The system shall provide IT staff with a dashboard to view, filter and update all open tickets.

Acceptance criteria:

1. Given an IT agent is logged in, when they open the dashboard, they see at least all tickets assigned to them or to their support group.

2. Given the dashboard is open, when the agent applies a filter (e.g. status = “Open”, category = “Network”), only matching tickets are displayed.
 3. Given the agent opens a ticket from the dashboard and changes its status, the updated status is saved and immediately visible in the dashboard list.
- The system shall support a searchable knowledge base accessible to both users and IT staff.

Acceptance criteria:

1. Given a user is logged in, when they navigate to the Knowledge Base and enter a search term, the system returns a list of relevant articles.
 2. Given the list of articles is displayed, when the user selects an article, the full content opens successfully and is readable.
- The system shall automatically send email notifications when a ticket is created, updated or resolved.
 - The system shall support ticket assignment and reassignment to specific IT agents or groups.

Non-Functional Requirements

- The portal shall be available at least 99% of the time during the academic year.
- Average page response time for common actions (view dashboard, submit ticket) shall be under 3 seconds under normal load.
- The user interface shall be usable on both desktop and mobile browsers.

Technical Requirements

- The system shall integrate with the university’s Single Sign-On (SSO) solution for authentication.
- The system shall support secure HTTPS communication and role-based access control for different user types.
- The system shall provide export or API access for reporting tools (e.g., CSV or REST API).

CRUD Table

Data Object	CRUD Action	Related Use Case / Functionality	Notes (Who / How)

Ticket	Create	Submit IT support request	Created by authenticated users (students, staff) via helpdesk portal.
Ticket	Read	View ticket details and status	Requester reads own tickets; IT staff read all tickets in their queues.
Ticket	Update	Add comments, change status, assign to agent / queue	IT staff update fields; requester can add clarifying comments.
Ticket	Delete	Close / archive resolved or invalid tickets	Only IT manager/admin can close/archive according to retention policy.
User	Read	Identify requester (name, role, department, contact details)	Read from central directory/SSO; no direct create/delete inside helpdesk.
KB Article	Create	Publish new knowledge base article from resolved issue	IT staff create and tag articles for self-service use.
KB Article	Read	Search and view solutions in the knowledge base	All authenticated users can read published articles.
KB Article	Update	Edit, improve, or deprecate existing article	Only authorised IT staff can update content and visibility.

Requirements Validation and Test Scenarios

Test Scenarios and Cases for Key Requirements

Test Case ID	Related Requirement	Test Steps	Expected Result
TC-001	The system shall allow users to submit IT tickets via a web portal, including category, description, urgency and attachments.	1. Log in as a student via SSO. 2. Open "New Ticket". 3. Select category "LMS access issue". 4. Enter description and set urgency to "High". 5. Attach a screenshot and click "Submit".	A new ticket is created with unique ID, category, description, urgency and attachment are stored and visible in "My Tickets".

TC-002	The system shall provide IT staff with a dashboard to view, filter and update all open tickets.	1. Log in as IT support agent. 2. Open the ticket dashboard. 3. Filter by category “Network” and status “Open”. 4. Open one ticket and change status to “In Progress”.	Dashboard initially shows only open “Network” tickets, selected ticket opens and its status is updated to “In Progress” and reflected in the list.
TC-003	The system shall support a searchable knowledge base accessible to both users and IT staff.	1. Log in as lecturer. 2. Go to “Knowledge Base”. 3. Search for “password reset LMS”. 4. Open one of the result articles.	The system returns relevant articles and allows the lecturer to open and read the full article without errors.
TC-004	The system shall automatically send email notifications when a ticket is created, updated or resolved.	1. As student, create a new ticket. 2. As IT agent, change its status from “Open” to “Resolved”. 3. Check the student’s email inbox.	The student receives an email after ticket creation (with ID and summary) and another email when the status changes to “Resolved”.
TC-005	The system shall support ticket assignment and reassignment to specific IT agents or groups.	1. Log in as helpdesk manager. 2. Open an existing ticket. 3. Assign it to group “Network Support”. 4. Log in as a member of “Network Support” and open the assigned tickets view.	The ticket’s assignee is updated to “Network Support” and appears in the assigned tickets list for a user in that group.

Conclusions and Recommendations

Summary of Findings

The analysis showed that the university’s current IT support is informal, fragmented, and not aligned with its digital-empowerment strategy. Requests are handled via emails, walk-ins, and phone calls, leading to lost tickets, long resolution times, lack of transparency, and no reliable performance metrics. Strategic, stakeholder, process, and

solution-design models (PESTLE, VMOST, POPIT, McKinsey 7S, CPPOLDAT, BAM) consistently highlighted the same core gaps: missing standard processes, no centralised system, poor data for decision-making, and low user satisfaction.

Proposed Solutions

The proposed Internal IT Helpdesk System introduces a centralized portal for ticketing, tracking, and knowledge management, supported by automation (routing, SLAs, notifications) and reporting. It standardises the end-to-end support process, clarifies roles (via RASCI), and improves information flows. The solution design and requirements specification define the key functional and non-functional capabilities needed to support efficient and transparent IT support for students, academic staff, and administrative staff.

Contribution to Strategic Goals

The helpdesk system directly supports the university's strategic aims of becoming a digitally empowered and student-centred institution. By reducing resolution times, increasing transparency, and enabling self-service, it improves student experience, protects teaching and research continuity, and optimises IT resource use. The Value Analysis (RAID, Benefit Dependency Network, cost/payback view) demonstrates that expected benefits in time savings, user satisfaction, governance, and risk reduction justify the investment and provide a clear rational basis for implementation.

References

PESTLE

- CIPD – what is PESTLE analysis:
<https://www.cipd.org/en/knowledge/factsheets/pestle-analysis-factsheet/>
- University of Sydney Library – explanation of PESTLE:
<https://library.sydney.edu.au/support/searching/pestle-analysis>
- PESTLE in education:
<https://pestleanalysis.com/pestle-analysis-of-the-education-industry/>

IT Helpdesk / IT Service Management

- What is ITSM:
<https://www.ibm.com/think/topics/it-service-management>
- ServiceNow – IT Service Management (example of helpdesk/ITSM solution):
<https://www.servicenow.com/products/it-service-management.html>

McKinsey 7S / Stakeholders / RACI

- McKinsey 7S framework:
<https://managementconsulted.com/mckinsey-frameworks/>
- Stakeholder power/interest matrix:
<https://simplystakeholders.com/stakeholder-matrix/>
- RACI / RASCI responsibility matrix:
<https://solutionshub.epam.com/blog/post/rasci-matrix>

Requirements & Testing

- ISTQB – test cases, test design:
<https://istqbinformation.wordpress.com/2016/12/23/chapter-4-test-design-techniques/>